parameters ε and `minPoints` such that all clusters will be partitioned correctly: figure 12.13 shows an example, illustrating how the result produced by DBSCAN is sensitive to the choice of parameters, and figure 12.14 shows another example with areas of different density that make it impossible to choose a value for epsilon that will cluster both areas properly;

- Related to this aspect, one of the problems with successfully running this algorithm is that it can be challenging to find the best values for its parameters when there is no previous knowledge on the dataset.

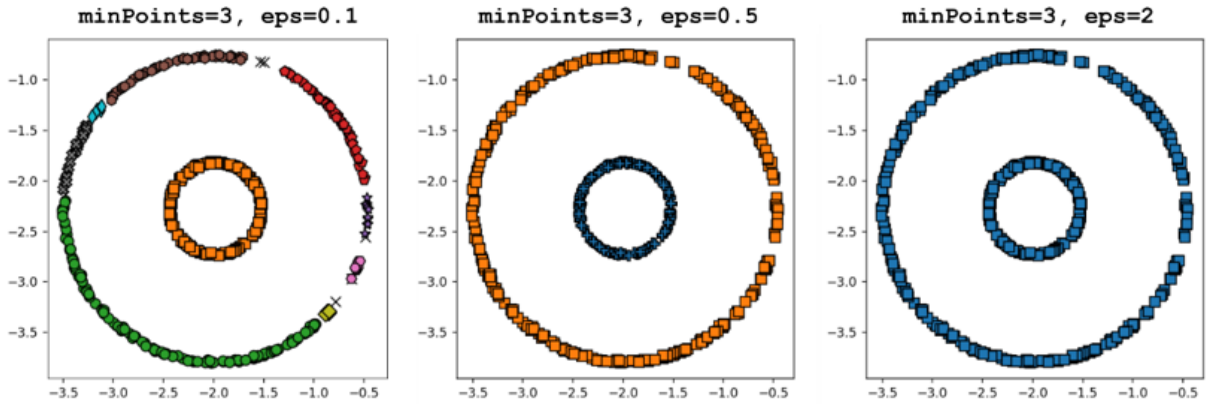


Figure 12.13 Using the first example dataset in figure 12.12, we can show how sensitive to the choice of parameters (in particular of ε) the result of DBSCAN is. All the examples ran DBSCAN on the same dataset with parameter `minPoints` set to 3 (the dimension of the domain plus 1).
(left) Using a value too small for ε causes the dense area to be too small, so data is partitioned into too many small clusters. A few points, plotted with X-shaped markers, are even marked as outliers.
(center) When the right value for the domain is